

Dr. Vipul Silwal

Mailing Address

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Education

PhD in Geophysics 2012-2018
University of Alaska Fairbanks, USA
Thesis: Earthquake source mechanisms and three-dimensional wavefield simulations in Alaska
Advisor: Dr. Carl Tape

Integrated BS and MS, Exploration Geophysics 2007-2012
Indian Institute of Technology (IIT), Kharagpur, India
Thesis: Moment Tensor Inversion in Alaska using Body and Surface Waves
Advisor: Dr. William Mohanty and Dr. Carl Tape

Experience

Assistant Professor

Department of Earth Sciences,
Indian Institute of Technology Roorkee, India June 2019 - present

Joint Faculty

International Center of Excellence for Dams,
Indian Institute of Technology Roorkee, India June 2023 - present

Research Associate

Department of Civil Engineering
Indian Institute of Technology Hyderabad, India Mar 2019 - May 2019

- Initiating and formalizing steps for seismicity and tectonic study in Himalayas
- Setting up HPC desktop for moment tensor inversion
- Meshing and three-dimensional simulations in northwestern Himalayas

Research Assistant

University of Alaska Fairbanks, USA 2012-2018

- Moment tensor inversion of earthquakes and VLFs in Alaska
- Estimation of uncertainty in moment tensors using probabilistic approach.
- Wavefield simulations in Wrangell, southern and interior Alaska.
- Preparing a reference velocity model of Alaska for tomographic inversion.

Field Technician

Alaska Earthquake Center, Alaska, USA Summer 2012, 2013

- Installation and maintenance of seismic stations in Alaska.

Research Assistant

IIT Kharagpur, India 2011-2012

- Computation of green's function and synthetics required for moment tensor inversion.

Research Intern

University of Tromsø, Norway Summer 2010

- Seismicity analysis of Hakon Mosbey Mud Volcano (Barents Sea).
- Offshore field trip in Barents Sea for sedimentary coring and seismic reflection survey.

Teaching

IITR

1. ESN-421 Numerical modeling in Geosciences (Autumn)
2. ESN-510 Numerical techniques and computer programming (Autumn)
3. ESN-102 Global Geophysics (Autumn)
4. ESN-422 Geophysical Inversion (Spring)
5. ESN-472 Advanced Seismology (Spring)
6. DSR-505 Dam Safety Surveillance, Instrumentation and Monitoring (Spring)

PhD Students

1. Rinku Mahanta - Waveform-based Moment tensor inversion in Himalayas (ongoing)
2. Rahul - 3D wavefield simulations and seismic imaging in Himalayas (ongoing)
3. Nivika Singh - Waveform-based source location of icequakes (ongoing)

UAF

Teaching Assistant, Physics 211

Spring 2015

- Teaching Physics 211 lab, Homework help

Student mentor

QingPing Yu, undergrad project on moment tensor inversion in interior Alaska.

Joshua Purba, undergrad project on moment tensor inversion in southern Alaska. (*now at University of Calgary*)

Funded Project (Research/ Consultancy)

1. Seismic imaging of central Himalayas using spectral element methodology, DST- SERB-SRG 2021-2024
2. Moment tensor inversion and three-dimensional wavefield simulation in Himalayas, IIT-FIG 2020-2023
3. Bihar 1934 and Uttarkashi 1991 Earthquake scenario Multi-State risk preparedness campaign, NDMA 2020
4. Geological and Geophysical investigation of Bageshwar Mines, Geocoin Ltd. 2022
5. Ultrasonic Seismic imaging application to Sandalwood trees, Dharmapal Satayapal Ltd., 2023-2024
6. Development of low-cost geophone sensors, iHUB-IITR 2022

Publications

1. Singh, SP, Padhy, S, and **V Silwal**, (Accepted), A high-resolution 3D shear velocity model for Cameroon using ambient noise tomography: Constraints from the CPSO algorithm, GJI-S-24-0346.
2. **Silwal, V**, Dwivedi, A , Mahanta, R, and H Mittal (In review) Nov 3, 2023 Nepal earthquake: Source mechanisms and structural response of Indo-Gangetic basin, Scientific Report.
3. Mahanta, R, **Silwal V**, Sharma, ML, and Mittal H, (In review) Moment tensor inversion and uncertainty analysis of earthquakes in Uttarakhand Himalayas, GJI-24-0368.
4. Mahanta, R., **Silwal, V.**, Sharma, M.L. (2024). Body Waves– and Surface Waves–Derived Moment Tensor Catalog for Garhwal-Kumaon Himalayas. In: Kumar, R., Singh, R., Kanhaiya, S., Maurya, S.P. (eds) Recent Developments in Earthquake Seismology. Springer, Cham. <https://doi.org/10.1007/978-3-031-47538-2-4>
5. Singh, SP and **V Silwal**, 2023, Enhanced crustal and intermediate seismicity in the Hindu Kush-Pamir region revealed by attentive deep learning model, Artificial Intelligence in Geosciences, Volume 4, Pages 150-163.

6. Kumar P, **Silwal V**, Mahanta R, Maurya VK, Kamal, Sharma ML, Ammani A, 2023. Near Real-Time Detection and Moment Tensor Inversion of the 11 May 2022, Dharchula Earthquake. *GeoHazards*.vol 4(4):515-525. <https://doi.org/10.3390/geohazards4040029>.
7. Kumar V, **V Silwal**, S Ray, 2023, Simulation of Seismic Wave Propagation Using Spectral-Element Method for Uttarakhand Himalayan Region with SPECFEM3D. 2nd EAGE/Aqua Foundation Indian Near Surface Geophysics Conference and Exhibition.
8. **V. Silwal**, C. Tape, and A. Lomax, 2018, Crustal earthquakes in the Cook Inlet and Susitna regions, southern Alaska, *Tectonophysics*, Vol. 745, doi: 10.1016/j.tecto.2018.08.013.
9. Alvizuri, C., **V. Silwal**, L. Krischer, and C. Tape, 2018, Estimation of full moment tensors including uncertainties for earthquakes, volcanic events and nuclear explosions, *Journal of Geophysical Research: Solid Earth.*, Vol. 123, doi: 10.1029/2017JB015325.
10. Tape, C., S. Holtkamp, **V. Silwal**, Y. Kaneko, J. Hawthorne, J. P. Ampuero, N. Ruppert, K. Smith, and M. E. West, 2018, Earthquake nucleation and fault slip complexity in the lower crust of central Alaska, *Nature Geoscience*, 11, doi: 10.1038/s41561-018-0144-2.
11. Tape, C., A. Lomax, **V. Silwal**, J. D. Agnew and B. Brettschneider, 2017, The 1904 Ms 7.3 Earthquake in Central Alaska, *Bulletin of the Seismological Society of America*, Vol. 107, No. 3, pp. 1147–1174, June 2017, doi: 10.1785/0120160178.
12. **V. Silwal** and C. Tape, 2016, Seismic moment tensors and estimated uncertainties in southern Alaska, *Journal of Geophysical Research: Solid Earth.*, v. 121, doi: 10.1002/2015JB012588.
13. C. Tape, **V. Silwal**, C. Ji, L. Keyson, M.E. West, and N. Ruppert, 2015, Transtensional tectonics of the Minto Flats fault zone and Nenana basin, central Alaska, *Bulletin of the Seismological Society of America*, Vol. 105, No. 4, pp. 2081–2100, August 2015, doi: 10.1785/0120150055.
14. C. Tape, M. West, **V. Silwal**, and N. Ruppert, 2013, Earthquake nucleation and triggering on an optimally oriented fault, *Earth and Planetary Science Letters*, v. 363, p. 231- 241. doi: 10.1016/j.epsl.2012.11.060.
15. D. Trivedi, K. Devi, I. Buynevich, P. Srinivasan, K. Ravisankar, **V. Silwal**, D. Sengupta, and R. R. Nair., 2012, Interpretation of Dune Genesis from the Sedimentological Data and Ground Penetrating Radar (GPR) Signatures: A case study from Ashirmata Dune Field, Mandvi Beach, Gujarat, India, *International Journal of Geosciences*, 2012, 3, 772-779, doi: 10.4236/ijg.2012.34078.

Reports/ Catalogs

1. Mahanta, R., **V. Silwal**, (2024), Processing local seismological waveform datasets for moment tensor inversion using mtuq. Zenodo. <https://doi.org/10.5281/zenodo.10499910>
2. Rinku, M., and **Silwal, V.** (2024). Moment tensor inversion and uncertainty analysis for 40 Uttarakhand Earthquakes (2010-2022). Zenodo. <https://doi.org/10.5281/zenodo.12541669>
3. **V. Silwal**. Seismic moment tensor catalog for crustal earthquakes in the Cook Inlet and Susitna region of southern Alaska, 2018. ScholarWorks@UA at <http://hdl.handle.net/11122/8383>: descriptor file, text file of catalog, figures with waveform fits, and input weight files.
4. A. Lomax, **V. Silwal**, and C. Tape. Hypocenter estimation for 14 earthquakes in south-central Alaska (1929–1975), 2018. ScholarWorks@UA at <http://hdl.handle.net/11122/8380>: descriptor file and zipped set of text files for each earthquake.
5. **V. Silwal**. Seismic moment tensors for six events in the Minto Flats fault zone, 2012–2016, 2018. ScholarWorks@UA at <http://hdl.handle.net/11122/8253>: descriptor file, text file of catalog, figures with waveform fits, and input weight files.
6. C. Tape, **V. Silwal**, and S. Holtkamp. Step-response signals recorded during earthquakes in Alaska, 2017. ScholarWorks@UA at <http://hdl.handle.net/11122/7947> (last accessed 2017-10-27).

7. **V. Silwal**. Seismic moment tensor catalog for Minto Flats fault zone (2000–2014), 2015. ScholarWorks@UA at <http://hdl.handle.net/11122/6267> (last accessed 2016-01-22): descriptor file, text file of catalog, composite figures of waveform fits and depth searches.
8. **V. Silwal**. Seismic moment tensor catalog for southern Alaska, 2015. ScholarWorks@UA at <http://hdl.handle.net/11122/6025> (last accessed 2016-01-22): descriptor file, text file of catalog, figures with waveform fits, depth searches, and uncertainty analyses.

Conferences/ Abstracts

1. Rahul, and **V. Silwal**, Effect of topography on synthetic waveforms for the Uttarakhand Himalayas, GSW Conference, BHU Varanasi, GSW78522, Mar 6-7, 2024.
2. Mahanta, R., **V. Silwal**, and Rahul, Comparative Analysis of Moment Tensor Estimation Methods for Seismic Source Mechanism Characterization, GSW Conference, BHU Varanasi, GSW65089, Mar 6-7, 2024.
3. Mishra, S. and **Silwal, V.**, Web-Based Active Fault Interactive Database: An inventory of Faults and Earthquakes of the Indian Subcontinent, vol. 2022, Art. no. IN41B-01, 2022.
4. Kharita, A. and **Silwal, V.**, SPEAQ - A Python Toolbox for Assessing Station Performance After an Event, vol. 2021, Art. no. S25E-0291, 2021.
5. Modrak, R., Tape, C., Alvizuri, C. R., **Silwal, V.**, Thurin, J., and Cleveland, M., Force and moment tensor uncertainty quantification with 1D and 3D Earth models, vol. 2020, Art. no.S045-0004, 2020.
6. B. Chow, Y. Kaneko, **V. Silwal**, and C. Tape. Adjoint tomography of the Hikurangi subduction zone and New Zealand's North Island. 2018. Abstract T51I-0282 presented at 2018 Fall Meeting, AGU, Washington, D.C., 10-14 Dec.
7. Modrak, R., Silwal, V., Alvizuri, C. R., and Tape, C., Moment tensor estimation and uncertainty quantification using mtuq, instaseis, obspy and pymc, vol. 2018, Art. no. S53E-0458, 2018.
8. R. Modrak, **V. Silwal**, C. Alvuzuri, and C. Tape. Moment tensor inversion and uncertainty quantification using mtuq, instaseis, obspy and pymc. 2018. Abstract S53E-0458 presented at 2018 Fall Meeting, AGU, Washington, D.C., 10-14 Dec.
9. C. Tape, S. Holtkamp, **V. Silwal**, J. Hawthorne, Y. Kaneko, J. P. Ampuero, C. Ji, N. Ruppert, K. Smith, and M. E. West. Earthquake nucleation and fault slip complexity in the lower crust of central Alaska. 2018. Abstract T43B-03 presented at 2018 Fall Meeting, AGU, Washington, D.C., 10-14 Dec.
10. C. Tape, C. Alvizuri, **V. Silwal**, and W. Tape. Uncertainties for seismic moment tensors and applications to nuclear explosions, volcanic events, and earthquakes. 2017. Abstract S44B-01 presented at 2017 Fall Meeting, AGU, New Orleans, La., 11-15 Dec.
11. C. Alvizuri, **V. Silwal**, L. Krischer, and C. Tape. Estimation of full moment tensors, including uncertainties, for earthquakes, volcanic events, and nuclear explosions. 2017. Abstract T2.3-P8 presented at the CTBT Science and Technology Conference 2017, Vienna, Austria.
12. **V. Silwal**, C. Tape, and E. Casarotti. Wavefield simulations of earthquakes in Alaska for tomographic inversion. 2017. Abstract S51D-0619 presented at 2017 Fall Meeting, AGU, New Orleans, La., 11-15 Dec.
13. **V. Silwal** and C. Tape. Crustal earthquakes in the Cook Inlet and Susitna regions, southern Alaska. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.

14. J. C. Purba, **V. Silwal**, and C. Tape. A 10-Year Catalog (2007–2016) of Seismic Moment Tensors in Southern Alaska. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.
15. C. Tape, S., Holtkamp, **V. Silwal**, J. Hawthorne, Y. Kaneko, J. P. Ampuero, N. Ruppert, K. Smith, and M. E. West. Slow-to-fast earthquake nucleation in the lower crust of central Alaska. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.
16. C. Tape, **V. Silwal**, and S. Holtkamp. An unwanted long-period step response: examples from Alaska earthquakes. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.
17. C. Tape, A. Lomax, **V. Silwal**, J. D. Agnew, and B. Brettschneider. The Ms7.3 earthquake of August 27, 1904 in central Alaska. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.
18. C. Alvizuri, **V. Silwal**, L. Krischer, and C. Tape. Estimation of full moment tensors, including uncertainties, for earthquakes, volcanic events, and nuclear tests. 2016. Abstract S31A-2701 presented at 2016 Fall Meeting, AGU, San Francisco, Calif., 12-16 Dec.
19. **V. Silwal** and C. Tape. Wavefield simulations of earthquakes in southern Alaska for tomographic inversion. 2016. Poster presented at 2016 CIG All Hands Meeting, Davis, Calif., 20-22 June.
20. C. Alvizuri, **V. Silwal**, and C. Tape. Estimation of full moment tensors with uncertainties. 2016. Poster presented at 2016 CIG All Hands Meeting, Davis, Calif., 20-22 June.
21. **V. Silwal** and C. Tape. Wavefield simulations of earthquakes in southern Alaska for tomographic inversion. 2016. Abstract presented at 2016 IRIS Workshop, Vancouver, Wash., 8-10 June.
22. C. Alvizuri, **V. Silwal**, and C. Tape. Estimation of full moment tensors with uncertainties. 2016. Abstract presented at 2016 IRIS Workshop, Vancouver, Wash., 8-10 June.
23. **V. Silwal** and C. Tape. Seismic moment tensors and estimated uncertainties in southern Alaska subduction zone. 2016. Abstract presented at 9th Biennial Workshop on Japan-Kamchatka-Alaska Subduction Processes, Fairbanks, Alaska, May 31 to June 3, 2016.
24. **V. Silwal** and C. Tape. Seismic moment tensors and estimated uncertainties in southern Alaska. 2016. Abstract presented at 2016 Alaska Geological Society Technical Conference, Fairbanks, Alaska, April 22.
25. C. Tape and **V. Silwal**. Earthquake source studies and seismic imaging in Alaska. 2015. Abstract T42B-05 presented at 2015 Fall Meeting, AGU, San Francisco, Calif., 14-18 Dec.
26. C. Tape, **V. Silwal**, C. Ji, L. Hutchinson, M. West, and N. Ruppert. Transtensional tectonics of the Minto Flats fault zone and Nenana basin, central Alaska. 2015. Abstract presented at the 2015 SSA Annual Meeting, Pasadena, California, April 21-23.
27. S. G. Holtkamp, N. A. Ruppert, **V. Silwal**, D. Christensen, and C. Nye. Recurring Swarms of Deep Long Period Earthquakes in the Denali Volcanic Gap Suggest a Continuation of Volcanic Processes in the Absence of Active Volcanism. 2014. Abstract presented at the 2014 AGU Annual Meeting, San Francisco, California, Dec 15-19.
28. **V. Silwal** and C. Tape. Seismic moment tensors in southern Alaska derived from body waves and surface waves. 2014. Abstract presented at the 2014 SSA Annual Meeting, Anchorage, Alaska, April 30 - May 2.

29. L. Keyson, **V. Silwal**, M. West, and C. Tape. Earthquakes of the Minto Flats seismic zone, central Alaska. 2014. Abstract presented at the 2014 SSA Annual Meeting, Anchorage, Alaska, April 30 - May 2.
30. **V. Silwal** and C. Tape. Seismic moment tensor inversion with posterior samples and uncertainties. 2013. Abstract S51A-2311 presented at 2013 Fall Meeting, AGU, San Francisco, Calif., 9-13 Dec.
31. **V. Silwal** and C. Tape. Seismic moment tensor inversion: Sampling the posterior probability distribution. 2013. Abstract presented at the 2013 CIG-QUEST-IRIS Workshop on Seismic Imaging of Structure and Source, July 14-17, 2013, Fairbanks, Alaska, USA.
32. C. Tape, M. West, **V. Silwal**, and N. Ruppert. Earthquake nucleation and triggering on an optimally oriented fault. 2013. Abstract presented at the 2013 CIG-QUEST-IRIS Workshop on Seismic Imaging of Structure and Source, July 14-17, 2013, Fairbanks, Alaska, USA.
33. C. Tape and **V. Silwal**. A three-dimensional seismic velocity reference model for Alaska. 2013. Abstract presented at 2013 EarthScope National Meeting, Raleigh, North Carolina, May 12-15.
34. C. Tape, M. West, **V. Silwal**, and N. Ruppert. Earthquake nucleation and triggering on an optimally oriented fault. 2013. Abstract presented at 2013 EarthScope National Meeting, Raleigh, North Carolina, May 12-15.
35. C. Alvizuri, C. Tape, **V. Silwal**, D. Christensen, M. West, and S. McNutt. Mecanismos focales de ondas P en el volcan Uturuncu. 2013. Poster presented at Workshop Centenario del Observatorio San Calixto, La Paz, Bolivia, April 29 to May 1.
36. J. Mienert, S. Buenz, B. Ferre, G. Dutta, and **V. Silwal**. Long-Term Seismological Observations of Eruptions of the Hakon Mosby Mud Volcano at the Barents-Sea Continental Margin (ESONET-LOOME Demonstration Mission). 2011. Poster presented at EGU General Assembly, Vienna, Austria, April 3-8.

Awards

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|---|-----------|
| 1. Brian R. Zelanka Scholarship for Best Graduate Student | 2018 |
| 2. Earthscope National Meeting Scholarship | 2017 |
| 3. Best student poster at Alaska Geological Society Conference | 2016 |
| 4. Geophysical Society of Alaska scholarship | 2015 |
| 5. SEG scholarship | 2014 |
| 6. IRIS travel grant for IRIS Workshop | 2014 |
| 7. Geophysical Institute travel grant for AGU Fall Meeting | 2013 |
| 8. INSPIRE scholarship by the Department of Science & Technology, India | 2007-2012 |

Computer Skills

Languages: Python, C, Fortran, Perl, Latex, Shell scripting
Software: MATLAB, Obspy, SPECFEM3D, GEOCUBIT, GMT, SAC
Platforms: Linux, Mac, Windows, CPU cluster, GPU cluster

Other Activities

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| 1. Seismology Seminar coordinator at Geophysical Institute, UAF | 2013-2014 |
| 2. TOTAL Well Log Analysis Course | 2011 |
| 3. Schlumberger PETREL Seismic Visualization and Interpretation Course | 2010 |

**Research
Statement**

I am interested in theoretical and computational aspects of seismology. Seismology is a data-rich science with tremendous opportunities for understanding source processes and Earth structure. To pursue this I perform forward modeling of wave propagation through a media representative of Earth structure. The misfit between the synthetic thus generated and the observed data is then minimized using different approaches. The inversion technique and the applied minimization depends on many factors, such as theory underneath, linear or non-linear problem, dimensionality of the problem, and computational resources available. The uncertainty in the solution is also an integral part. For source inversion I have tried to quantify the uncertainty using Bayesian approach. These earthquake sources are the priori for the tomographic inversion. With the recent advances in high performance computing people have been able to carry out adjoint tomography at both regional and global scale. Similar effort is undergoing by us to better understand structure of Alaska and Himalayas.